

A Variable OSR $\Delta\Sigma$ ADC with LSB-First SAR ADC as Quantizer

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Introduction

- Real world $\rightarrow$ Analog
- Processing $\rightarrow$ Digital
- ADC = Interface between the two

ADC performance parameters

- Resolution (accuracy)
- Bandwidth (speed)
- Power consumption (efficiency)

Literature Review: Flash ADC ¹

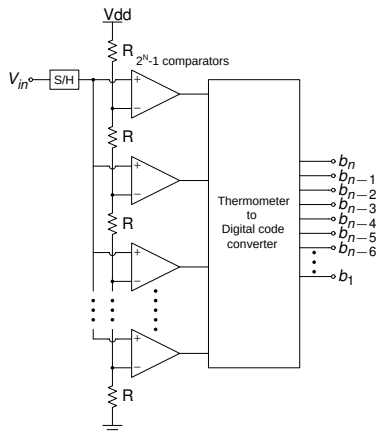


Figure: N-bit Flash ADC

¹B. Murmann, "ADC Performance Survey 1997–Present," Stanford University.

Literature Review: SAR ADC

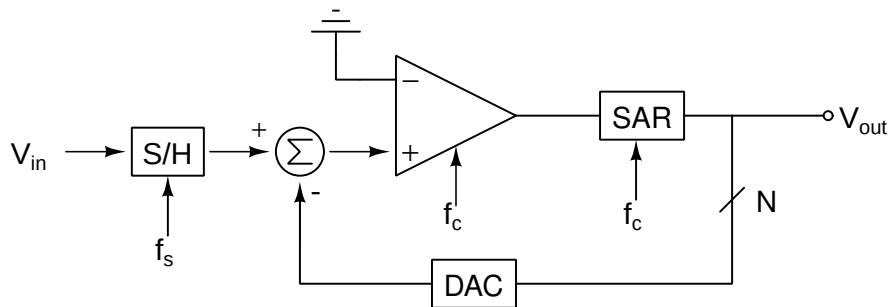


Figure: N-bit SAR ADC²

²Jacob Baker, *CMOS: Circuit Design, Layout, and Simulation*, 3rd ed., Wiley, 2010.

Literature Review: Delta-Sigma Modulator (DSM)

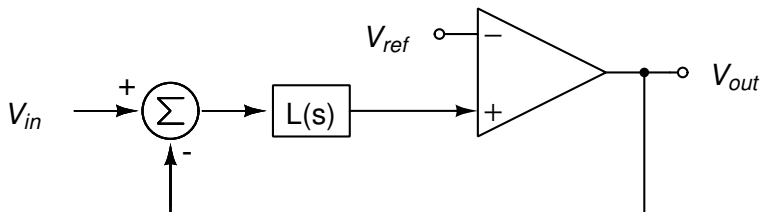
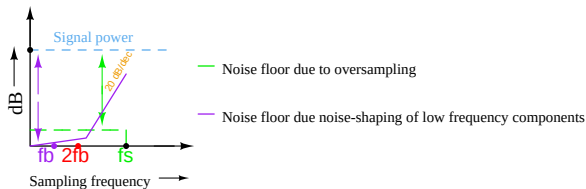


Figure: Delta-Sigma Modulator ³



³S. Pavan, R. Schreier, and G. Temes, *Understanding Delta-Sigma Data Converters*, 2nd ed., Wiley, 2017.

Literature Review: SAR ADC vs $\Delta\Sigma$ Modulator⁴

Nyquist-rate ADC

- $\text{SNR} = 6.02N + 1.76$

$\Delta\Sigma$ Modulator

- $\text{SNR} = 6.02(1) + 1.76 + 30 \log_{10} \left(\frac{f_s}{2f_{\text{in}}} \right) - C$

⁴B. Murmann, “ADC Performance Survey 1997–Present,” Stanford University. 

Literature Review: Noise-Shaping SAR ADC

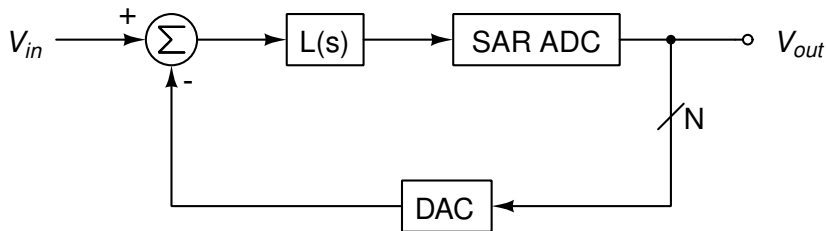


Figure: Noise-Shaping SAR ADC ⁵

For a N-bit 1st-order noise-shaping ADC,

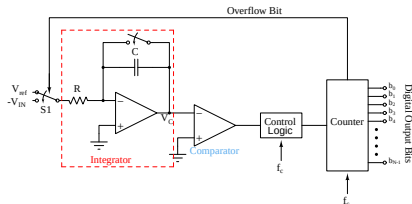
$$\text{SNR} = 6.02N + 1.76 + 30 \log_{10} \left(\frac{f_s}{2f_{in}} \right) - C_1$$

⁵S. Pavan, R. Schreier, and G. Temes, *Understanding Delta-Sigma Data Converters*, 2nd ed., Wiley, 2017.

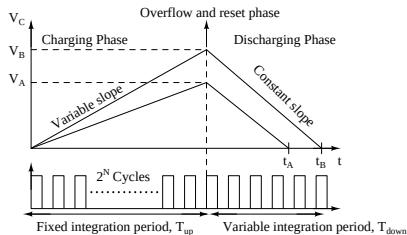
High Resolution ADCs

- Noise-Shaping SAR ADC - Noise shaping at lower frequencies
- Dual Slope ADC - Achieves high resolution using integration-based techniques.

Dual Slope ADC⁶



(a) Dual Slope ADC



(b) Working of Dual Slope ADC in different phases

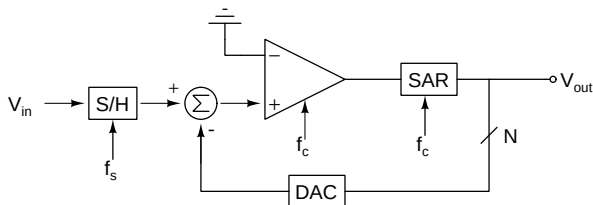
- Depends on time ratios for AD conversion.

- $$V_{out} = \frac{T_{down}}{T_{up}} V_{ref}$$

- Cycles per conversion - $2^N + 1$ to 2^{N+1}

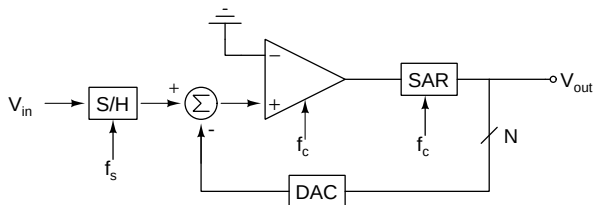
⁶Jacob Baker, *CMOS: Circuit Design, Layout, and Simulation*, 3rd ed., Wiley, 2010.

SAR ADC



- 1 Binary search algorithm.
- 2 N-bit ADC takes N cycles for each conversion.

SNR of ADC

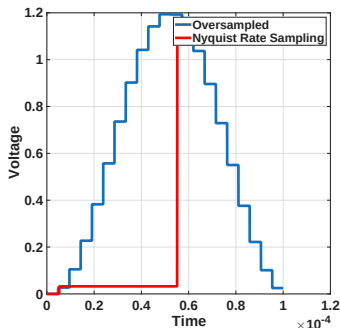


$$\text{SNR} = 6.02N + 1.76 + 10\log(f_s/(2 \times f_{in}))$$

For a N-bit SAR ADC,

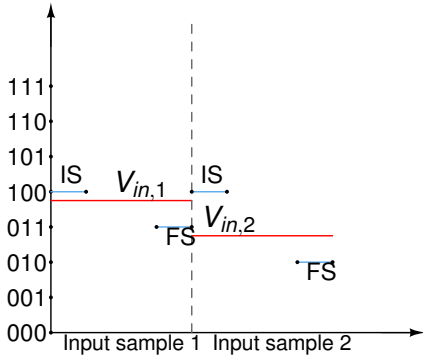
$$\text{SNR} = 6.02N + 1.76 + 10\log(f_c/(2 \times f_{in} \times k))$$

Number of cycles per conversion, $k=N$ (In conventional SAR ADC)

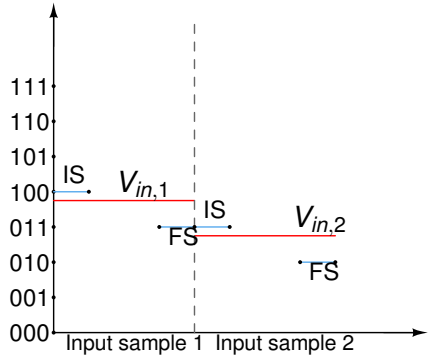


- 10 KHz single-tone sine with amplitude 1V - $\sin(2\pi \times 10000t)$
- Difference when sampled at 1 MHz - $\sin(2\pi \times \frac{10^4}{10^6}) = 62 \text{ mV}$
- For an 8-bit ADC, $62 \text{ mV} \approx 16 \text{ LSBs}$

Sine wave sampled at Nyquist rate
v Oversampled sine wave



(a)



(b)

Figure: Examples showing Initial state(IS) and Final state(FS) of (a)SAR ADC and (b)LSB-First SAR ADC for different sampled inputs, $V_{in,1}$ and $V_{in,2}$

LSB-First SAR ADC

- 2 to $2N+1$ cycles⁷.

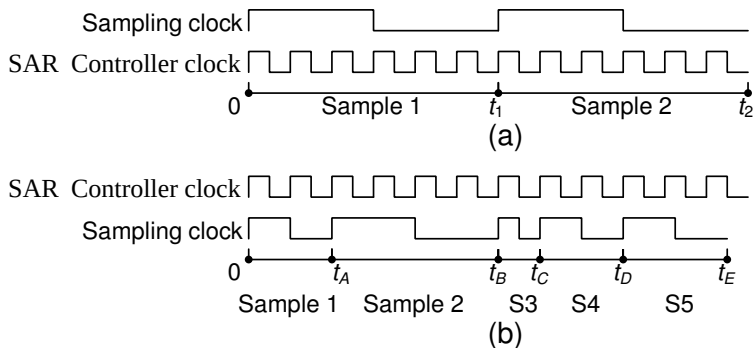


Figure: Example sampling clock for (a)SAR ADC and (b)Implemented ADC architecture

⁷F. M. Yaul and A. P. Chandrakasan, "A 10-bit SAR ADC with data-dependent energy reduction using LSB-first successive approximation," *IEEE J. Solid-State Circuits*, 2014.

Non-Uniform Sampling ADC

LSB-First SAR ADC : Working

Comparator output	Digital output	ADC operating phase
	1101000110	Initial state
Logic 1	1101000111	LSB phase
Logic 1	1101001111	LSB phase
Logic 1	1101011111	LSB phase
Logic 0(First transition)	1101010111	MSB phase
Logic 1	1101011011	MSB phase
Logic 0	1101011001	MSB phase
Logic 1	1101011010	MSB phase
Logic 1	1101011011	MSB phase
End of conversion	1101011011	Required output

LSB-First SAR: Implementaion

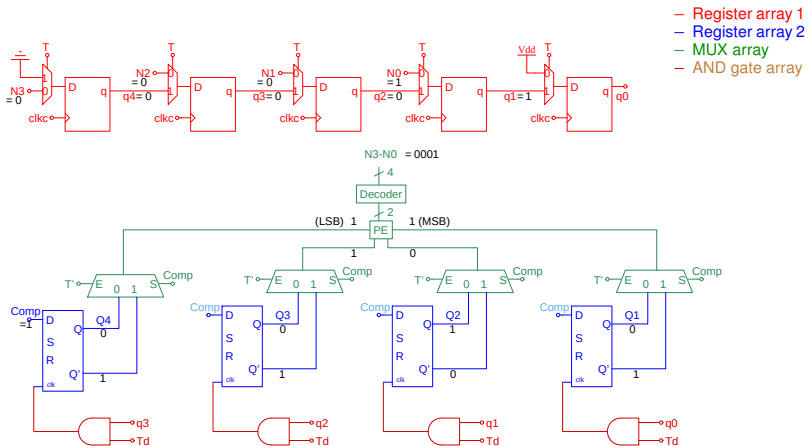


Figure: 4-bit Successive Approximation Register for LSB-First SAR ADC

SET and RESET signal generator

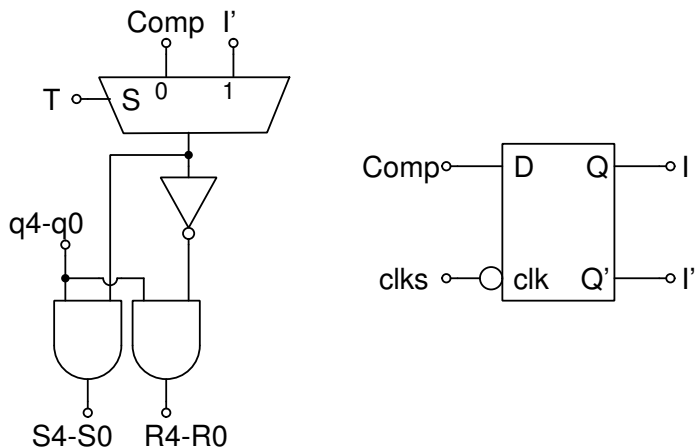


Figure: Digital logic to control set and reset operations of output registers

LSB-First SAR: Implementaion

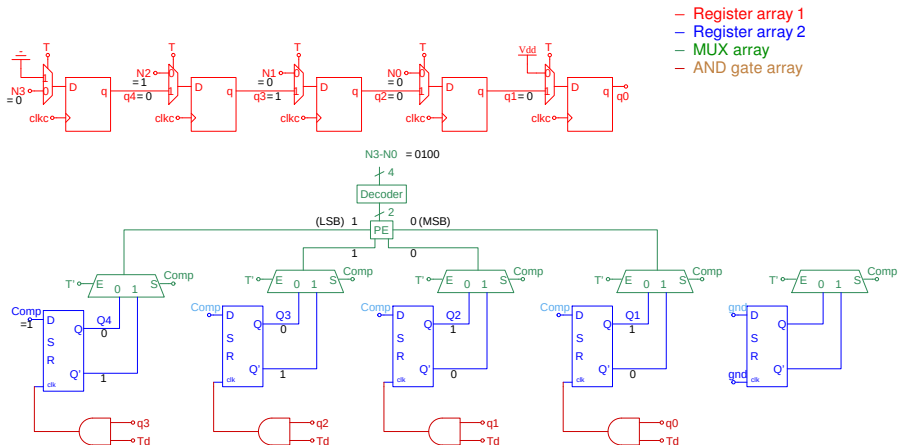


Figure: 4-bit Successive Approximation Register for LSB-First SAR ADC

LSB-First SAR ADC : Working

Comparator output	Digital output	ADC operating phase
	1101000110	Initial state
Logic 1	1101000111	LSB phase
Logic 1	1101001111	LSB phase
Logic 1	1101011111	LSB phase
Logic 0(First transition)	1101010111	MSB phase
Logic 1	1101011011	MSB phase
Logic 0	1101011001	MSB phase
Logic 1	1101011010	MSB phase
Logic 1	1101011011	MSB phase
End of conversion	1101011011	Required output

Transition Detector

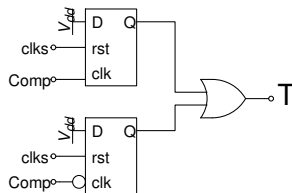


Figure: Transition detector

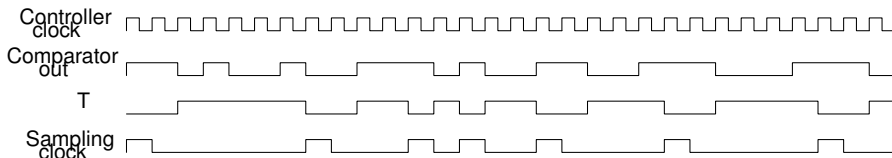


Figure: Example waveform showing working of Transition Detector

LSB-First SAR ADC : Working

Comparator output	Digital output	ADC operating phase
	1101000110	Initial state
Logic 1	1101000111	LSB phase
Logic 1	1101001111	LSB phase
Logic 1	1101011111	LSB phase
Logic 0(First transition)	1101010111	MSB phase
Logic 1	1101011011	MSB phase
Logic 0	1101011001	MSB phase
Logic 1	1101011010	MSB phase
Logic 1	1101011011	MSB phase
End of conversion	1101011011	Required output

Successive Approximation Register (SAR)

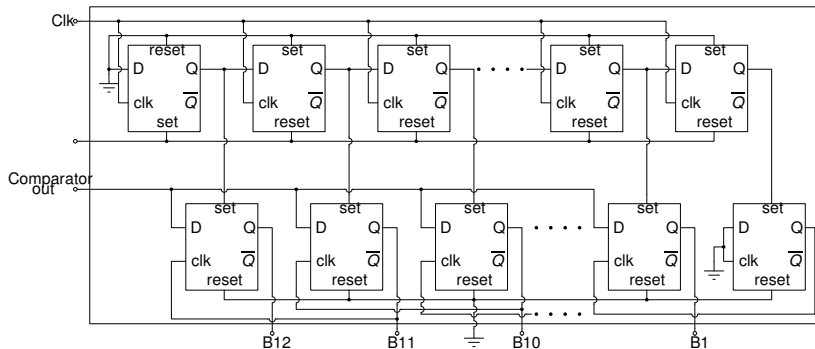


Figure: Successive Approximation Register for Conventional SAR ADC

LSB-First SAR: Implementaion

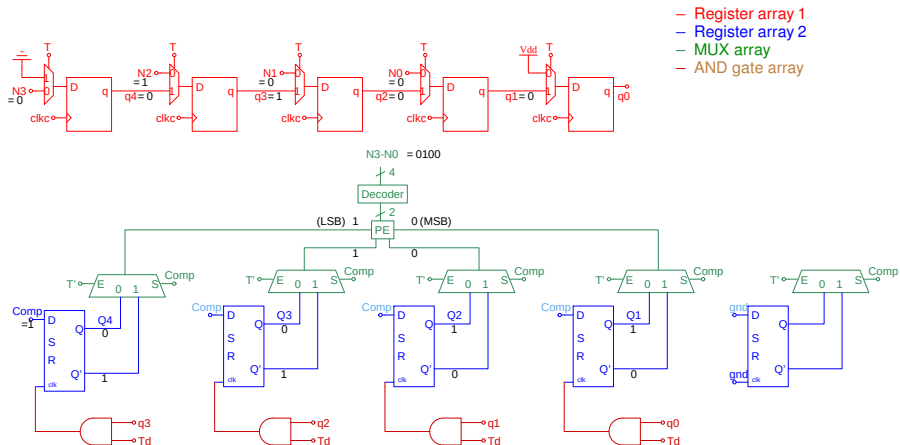


Figure: 4-bit Successive Approximation Register for LSB-First SAR ADC

LSB-First SAR ADC: Sampling clock generation

Comp out	ADC output	Operating phase	Bits activity tracker
	1101000110	Initial state	
Logic 1	1101000111	LSB phase	0000000001
Logic 1	1101001111	LSB phase	0000001000
Logic 1	1101011111	LSB phase	0000010000
Logic 0	1101010111	MSB phase	0000001000
Logic 1	1101011011	MSB phase	0000000100
Logic 0	1101011001	MSB phase	0000000010
Logic 1	1101011010	MSB phase	0000000001
Logic 1	1101011011	MSB phase	
EOC	1101011011	Required output	

Variable-rate SAR ADC

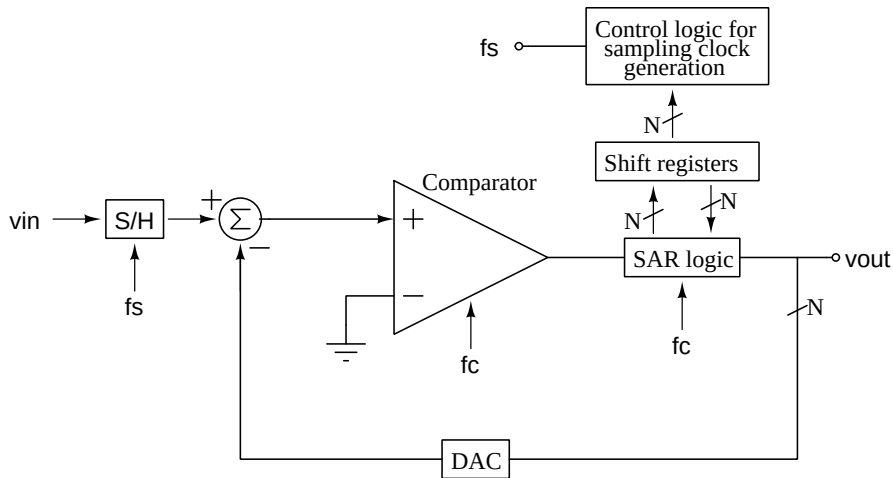


Figure: Block diagram of Non-Uniform Sampling SAR ADC

Observations

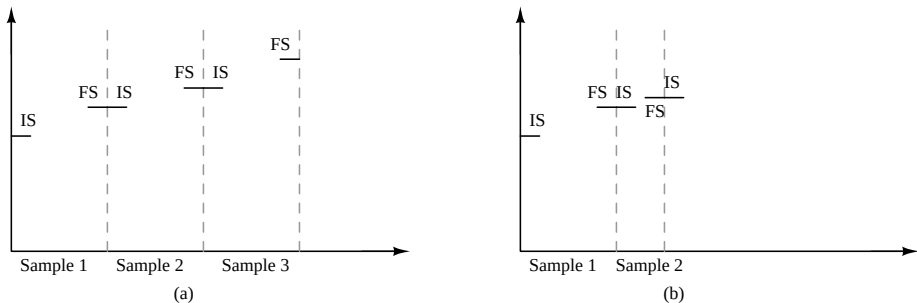
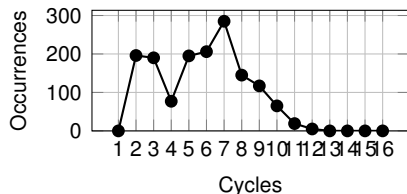


Figure: Examples showing Initial state(IS) and Final state(FS) of (a)LSB-First SAR ADC and (b)Variable-rate SAR ADC for different samples, Sample 1,2,3

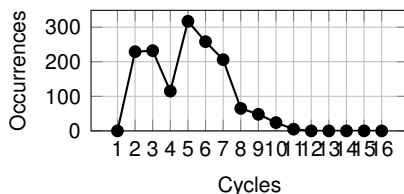
Observations: Cycle Count Distribution

SAR clock frequency = $f_c = 10\text{MHz}$

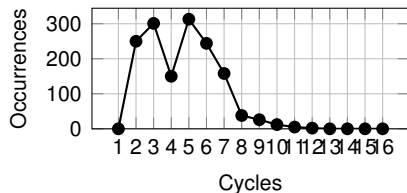
TB1 ($f_{in}=7812.5$)



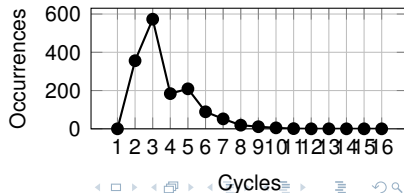
TB2 ($f_{in}=6250$)



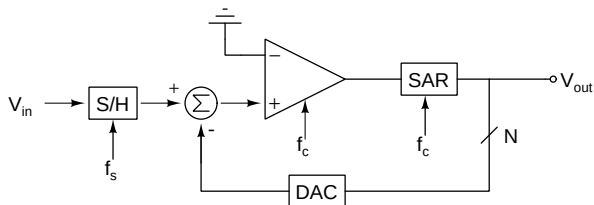
TB3 ($f_{in}=3125$)



TB4 ($f_{in}=2500$)

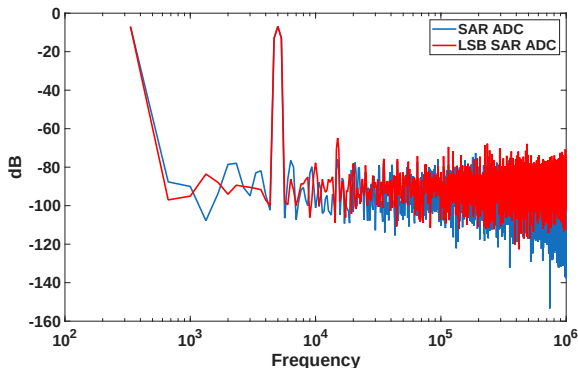


Observations



Test bench	INPUT SIGNALS		SAR ADC	LSB-FIRST SAR
Serial No.	f_c (in Hz)	f_{in} (in Hz)	ACPC	ACPC
TB1	10M	25000	8	6.55
TB2	10M	12500	8	5.3
TB3	10M	7812.5	8	4.49
TB4	10M	6250	8	4.12
TB5	10M	3125	8	3.5
TB6	10M	2500	8	3.2

Results



- Frequency of the input signal is 5KHz
- Frequency of the SAR clock signal is 16MHz.
- Average cycles per conversion = $k = 3.5$
- SNR improvement = 3dB

Results

- Frequency of the input signal is 5KHz
- Frequency of the SAR clock signal is 16MHz.
- Average cycles per conversion = $k = 3.5$
- Expected SNR improvement =
$$(SNR)_{LSB-SAR} - (SNR)_{Conv SAR} = 10 \log \left(\frac{k_{Conv SAR}}{k_{LSB SAR}} \right)$$
$$= 10 \log(8/3.5) = 3.59 \text{ dB}$$
- Observed SNR improvement = $76.5 \text{ dB} - 73 \text{ dB} = 3.5 \text{ dB}$

1st order DSM

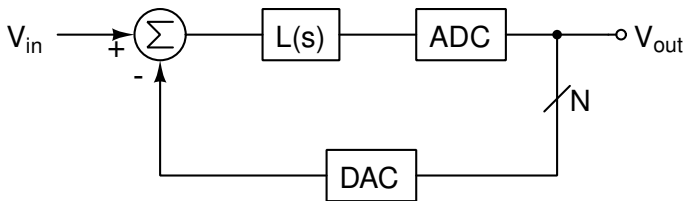
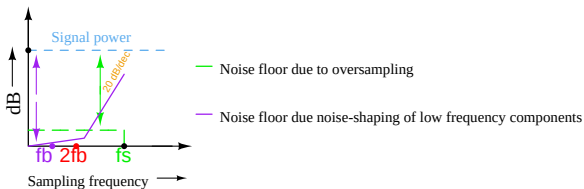
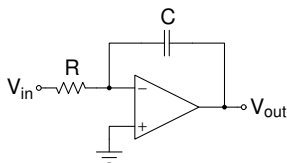


Figure: 1st order noise shaping ADC



Design of Loop filter



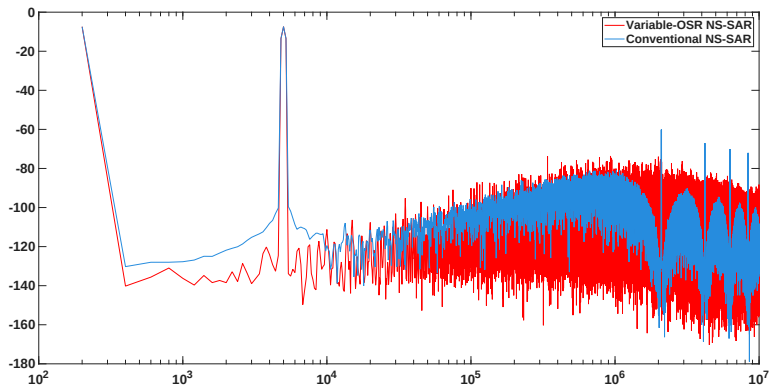
$$L(s) = \frac{-1}{sRC}$$

$$\frac{1}{RC} = \frac{f_c}{2 \times k_{avg}}$$

The Noise Transfer Function for the respective loop filter will be

$$NTF(z) = 1 - \frac{1}{RC(1 - e^{-T/RC}z^{-1})}$$

Results



- Frequency of the input signal is 5KHz
- Frequency of the SAR clock signal is 16MHz.
- Average cycles per conversion = $k = 3.5$

Results

- Frequency of the input signal is 5KHz
- Frequency of the SAR clock signal is 16MHz.
- Average cycles per conversion = $k = 3.5$
- Expected SNR improvement =
$$(\text{SNR})_{\text{NS LSB-SAR}} - (\text{SNR})_{\text{Conv NS SAR}} = 30 \log \left(\frac{k_{\text{Conv SAR}}}{k_{\text{LSB SAR}}} \right)$$
$$= 30 \log(8/3.5) = 10.77 \text{ dB}$$
- Observed SNR improvement = $116.4 \text{ dB} - 108.6 \text{ dB} = 7.8 \text{ dB}$

2nd order DSM

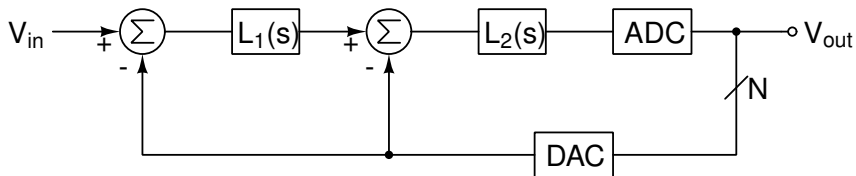
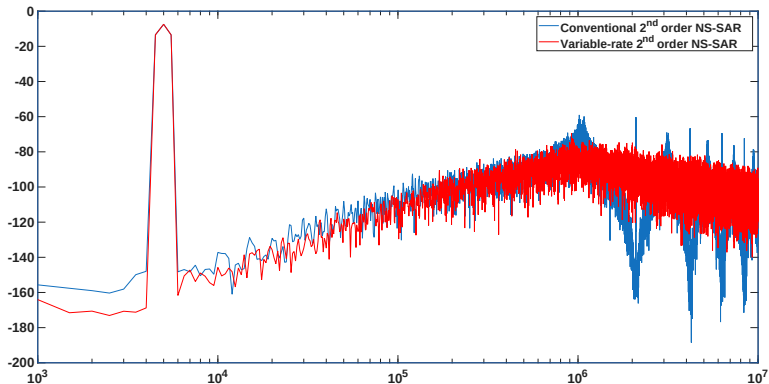


Figure: 2nd order noise shaping ADC

Results



- Frequency of the input signal is 5KHz
- Frequency of the SAR clock signal is 16MHz.
- Average cycles per conversion = 3.5

Results

- Frequency of the input signal is 5KHz
- Frequency of the SAR clock signal is 16MHz.
- Average cycles per conversion = $k = 3.5$
- Expected SNR improvement =
$$(SNR)_{LSB-SAR} - (SNR)_{Conv SAR} = 50 \log \left(\frac{k_{Conv SAR}}{k_{LSB SAR}} \right)$$
$$= 50 \log(8/3.5) = 17.95 \text{ dB}$$
- Observed SNR improvement = $156.6 \text{ dB} - 140.5 \text{ dB} = 16.1 \text{ dB}$

Future work: Decimator

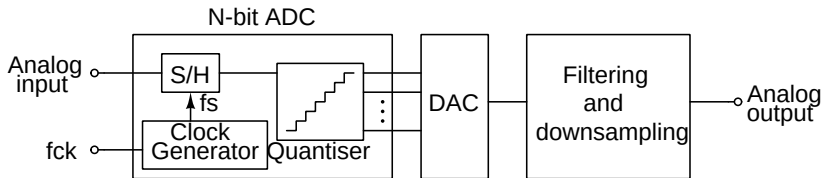


Figure: Quantization and decimation

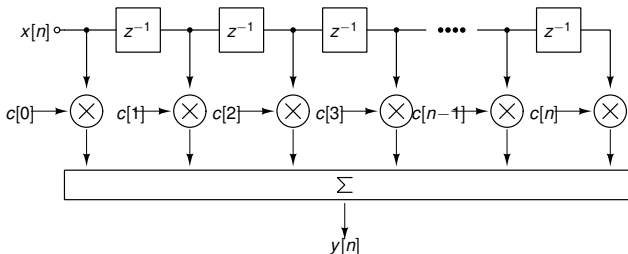


Figure: FIR structure

Future work: DAC design choices

- SAR ADC uses Charge-Sharing DAC in the feedback in the architecture.
- MSB capacitors stay in the same states for longer periods.
- Presence of parasitics lead to charge leakage, which will affect the ADC performance.
- A possible solution would be to reset the capacitors periodically.
- Explore the performance of the ADC with other DAC architectures.

Results and Conclusions

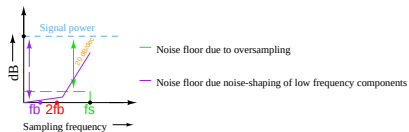
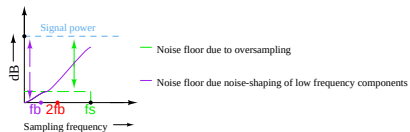
- This work proposes a non-uniform sampling technique to enhance the SNR of the ADC.
- Input signal frequency=5 KHz, ADC clock=16 MHz.

Architecture	Noise Shaping	Order	SNDR (dB)
SAR ADC	No	-	73dB
Variable OSR ADC	No	-	76.5dB
SAR ADC	Yes	1	108.6dB
Variable OSR ADC	Yes	1	116.4dB
SAR ADC	Yes	2	140.5dB
Variable OSR ADC	Yes	2	156.6dB

Thank you

Comments and Queries

● Why shaped noise show discontinuity in Fig2.17?

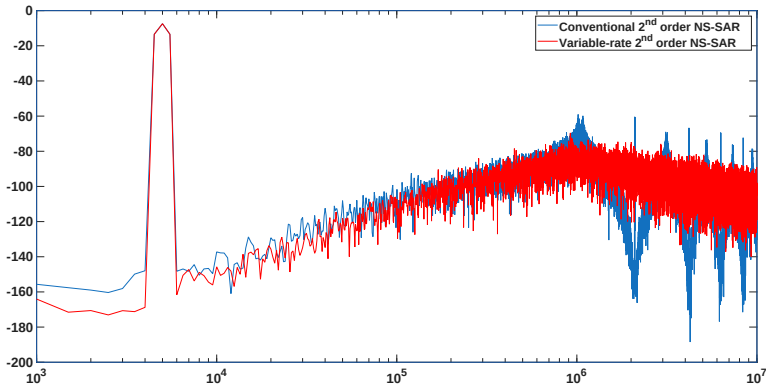


- What is the target resolution? What is achieved resolution? Is it the best noise shaped SAR architecture for this resolution? What is FoM of your design. What is the power dissipation? How is it when compared to other noise shaped SAR ADC architecture.
 - 1 The aim is to achieve better SNR without compromising the input signal bandwidth.
 - 2 Considering the achieved results for the reduction in the inband noise without affecting input signal bandwidth it can be concluded it the best noise shaped SAR architecture for this resolution.
 - 3 Choice of DAC plays a role in power dissipation.

- As the filtering is still active so how the power efficiency of SAR ADC architecture is maintained?
 - ① The implementation was done with an active filter as proof of concept.
 - ② It can be done with passive filter

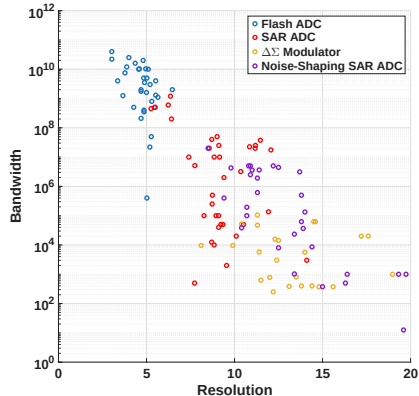
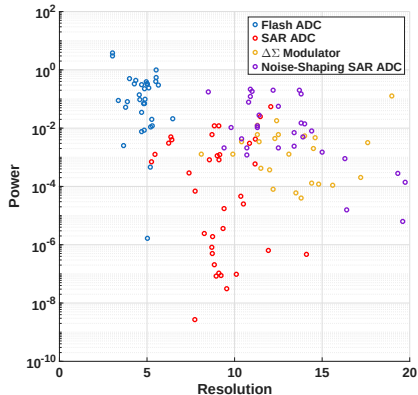
Q.4

- The Fig.4.13 has significant peaking in NTF? What is the reason behind it. How it affects the stability of the modulator?



ADC for Tire Pressure Monitoring System (TPMS)

Choice of ADC architecture



12-bit SAR ADC

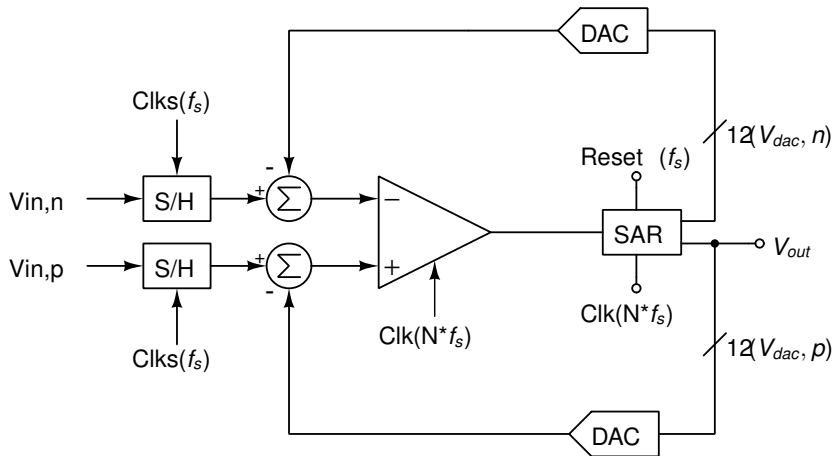


Figure: Block diagram of fully differential 12-bit SAR ADC

Comparator

- Architecture : Sense amplifier based comparator
- Worst case offset observed across PVT : 16 mV(Step size of ADC=366 μ V)
- Worst case offset observed across PVT after calibration : 0.35 mV

DAC

- Architecture : Split array Charge-sharing DAC

SAR

- Architecture : Sense amplifier based D-FF with latch modified to have set and reset pins

Results and Conclusion

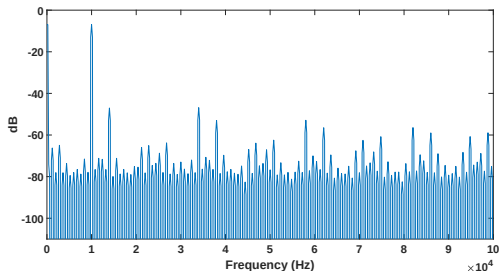


Figure: Output spectrum of 12-bit SAR ADC

Test Setup:

- Throughput=1 MHz
- SINAD @ 10 KHz = 70.39 dB
- ENOB = 11.44 bits

THANK YOU

12-bit SAR ADC

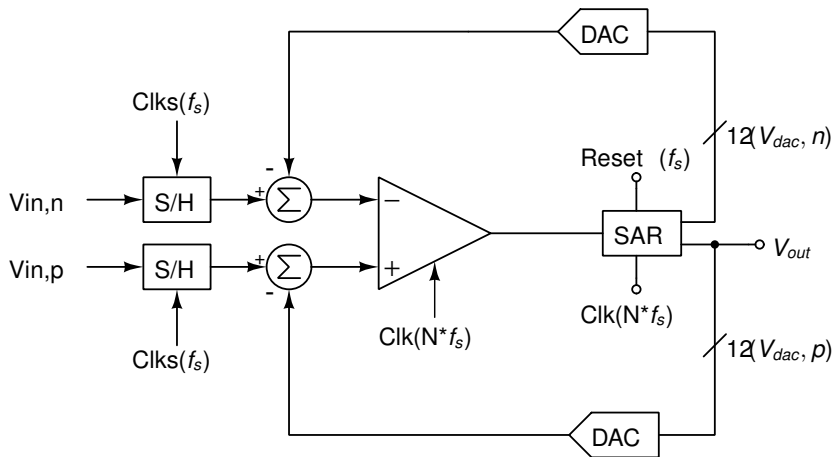
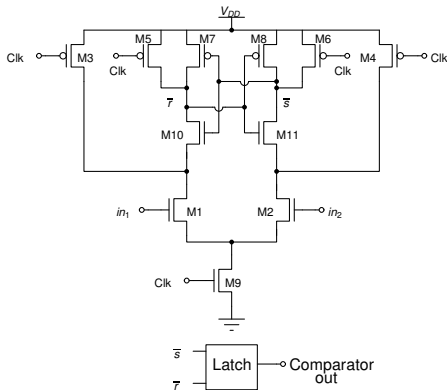
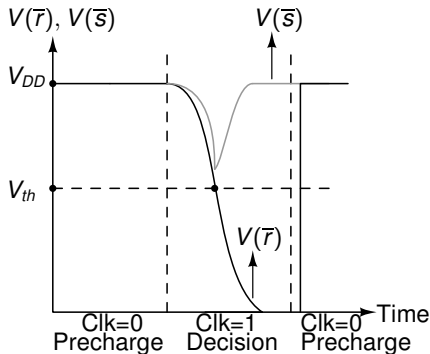


Figure: Block diagram of fully differential 12-bit SAR ADC

Design of Dynamic comparator⁸



(a) Sense Amplifier Comparator



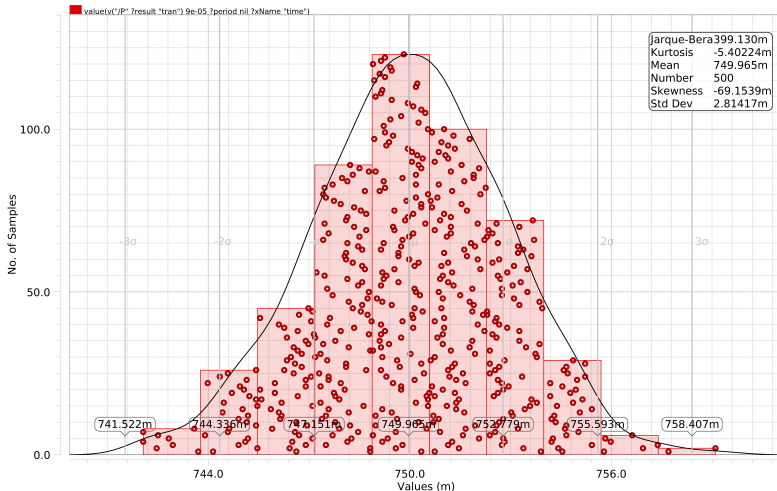
(b) Precharge and decision phases
($in_1 > in_2$)

⁸B. Razavi, "The design of a comparator," *IEEE Solid-State Circuits Magazine*, 2020.

Monte Carlo Simulation results: Before calibration

value(v("/P" ?result "tran") 9e-05 ?period nil ?xName "time")

1



Worst case offset ≈ 10 mV

Comparator calibration

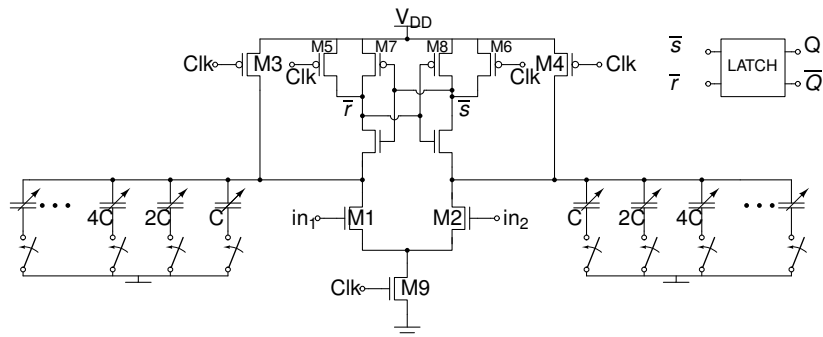
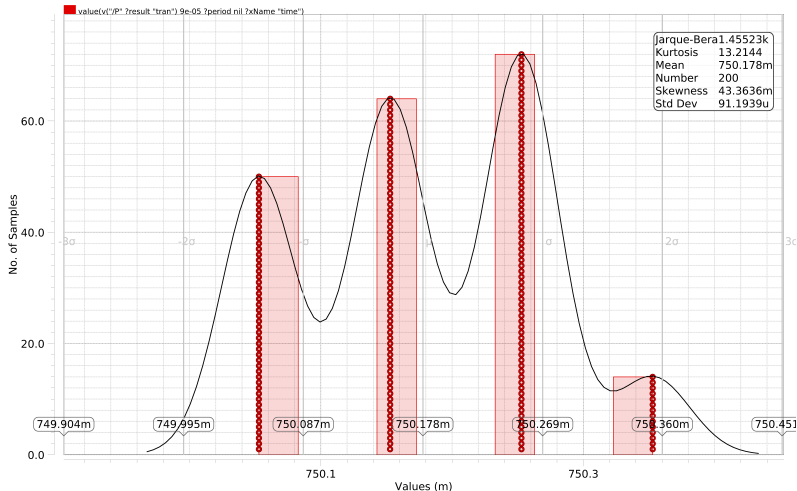


Figure: Calibration of Sense Amplifier comparator

Monte Carlo Simulation results: After calibration

value(v("/P" ?result "tran") 9e-05 ?period nil ?xName "time")

1



Worst case offset = 0.35 mV

Successive Approximation Register (SAR)

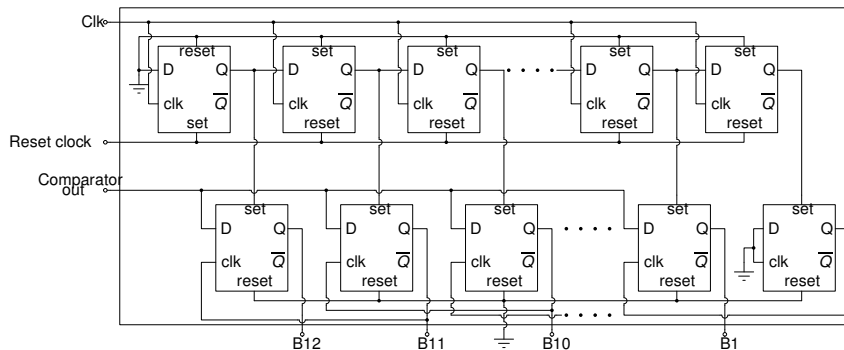
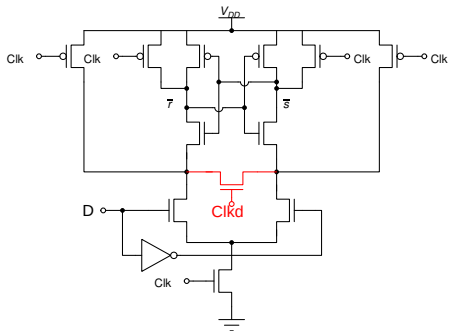


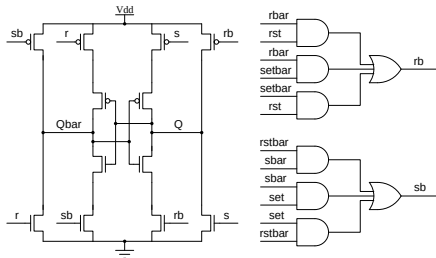
Figure: 12-bit successive approximation register⁹

⁹Jacob Baker, *CMOS: Circuit Design, Layout, and Simulation*, 3rd ed., Wiley, 2010.

Design of D-Flip Flop



(a) Sense Amplifier Comparator based
D-FF^a



(b) Latch

^aB. Razavi, "The design of a comparator," *IEEE Solid-State Circuits Magazine*, 2020.

Split-Array Charge sharing DAC

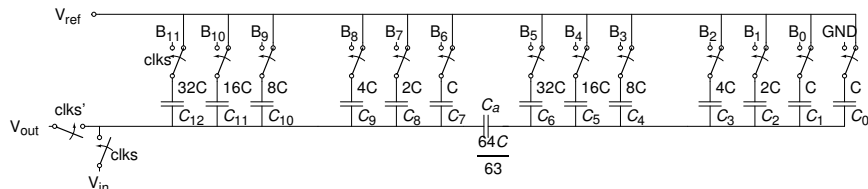


Figure: Split-Array CDAC schematic ¹⁰

¹⁰Jacob Baker, *CMOS: Circuit Design, Layout, and Simulation*, 3rd ed., Wiley, 2010.

Results and Conclusion

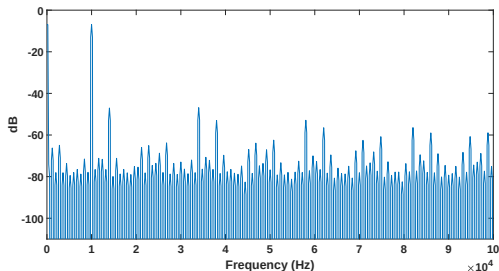
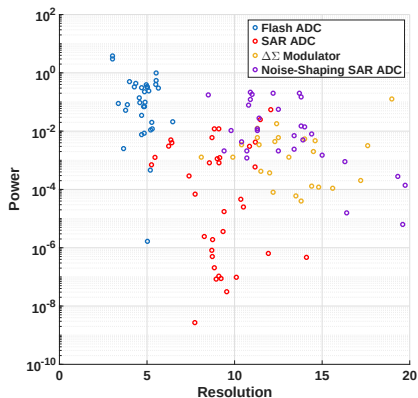


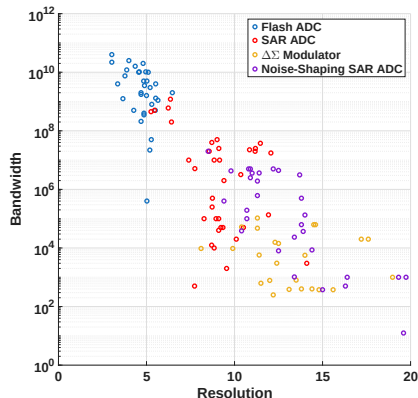
Figure: Output spectrum of 12-bit SAR ADC

Test Setup:

- Throughput=1 MHz
- SINAD @ 10 KHz = 70.39 dB
- ENOB = 11.44 bits



(a) Power vs Resolution



(b) Speed vs Resolution

¹¹B. Murmann, “ADC Performance Survey 1997–Present,” Stanford University.

